



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 06.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No	: CE G567
Course Title	: Highway Design
Instructor-in-Charge	: Nishant Bhargava
Instructor(s)	: --
Tutorial/Practical Instructors	: --

1. Course Description

Highway functional classification; route layout and selection, design controls and criteria: turning paths, driver performance, traffic characteristics; highway capacity; access control; safety; environment; Elements of design: sight distance, horizontal alignment, transition curves, super elevation and side friction; vertical alignment: - grades, crest and sag curves; highway cross-sectional elements and their design; at-grade Intersections - sight distance consideration and principles of design, canalization, mini roundabouts, layout of roundabouts, inter-changes: major and minor interchanges, entrance and exit ramps, acceleration and deceleration lanes, bicycle and pedestrian facility design; parking layout and design; terminal layout and design.

2. Scope and Objective of the Course

The highway vehicle and individual users are integral parts of transportation safety and efficiency. The objective of the course is to expose the students to the various components of the geometric design of highways.

3. Text Books

T1 : AASHTO-2001: A Policy on Geometric Design of Highways and Streets; American Association of State Highway and Transportation Officials.

Reference Books

- R1: Kadiyali, L. R. Traffic Engineering and Transport Planning, Khanna Publishers, latest Edition.
R2: Khanna S. K. and Justo C.E.G. Highway Engineering, Nem Chand and Bros, Latest Edition.
R3: Relevant Federal Highway Administration Reports and NCAT Reports
R4: Relevant IRC codes



4. Course Plan

Module No.	Lecture Session	Reference	Learning outcomes
1-2	Highway functional classification, functional system characteristics	T1, R1-R4	Functions of highways
3-4	Design vehicles, minimum turning path, driver performance	T1, R1-R4	Design controls and criteria
5-9	Volume, directional distribution, traffic composition, speed, highway capacity, levels of service	T1, R1-R4	Traffic characteristics
10-11	Walking speed, walkway capacity and bicycle facilities	T1, R1-R4	Facilities for pedestrians and bicycles
12-15	Stopping Sight distance, decision sight distance, passing sight distance	T1, R1-R4	Sight distance
16-22	Superelevation and its design consideration, transition curve, methods of attaining super-elevation, off-tracking and widening of roads	T1, R1-R4	Design of horizontal alignment
23-26	Critical length of grades, climbing lanes, design controls of vertical curves: crest and sag, climbing and emergency escape lane	T1, R1-R4	Design of vertical alignment
27-40	Types of roundabouts, channelizing islands, overpass and underpass and interchanges, entrance and exit ramps, acceleration and deceleration lanes, terminal layout and design.	T1, R1-R4	Design of at-grade and grade separated intersections
38-40	Bicycle and pedestrian facility design, parking layout and design.	T1, R3	Design of facilities for non-motorized vehicles, pedestrian movement and parking

5. Evaluation Scheme

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 min.	25	10/10/2026 AN2	CB/OB
Comprehensive Examination	3 hr.	35	16/12/2026 AN	CB/OB
Quizzes	45 min.	10	*	OB
Project	-	30	*	OB

* shall be announced prior



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

6. Chamber Consultation Hour

Wednesday 3 pm – 4 pm (1212 – F)

7. Notices

Notices of different components of evaluation scheme will be sent by email.

8. Grading Policy

- If the student absents herself/ himself even in one of the evaluative components (listed in the evaluation scheme) entirely, his / her performance shall be reported as NC (Not Cleared) regardless of his/ her final total score in the course (See Clause 4.19 of BITS Academic Regulations). Students gaining less than 10% overall, may be awarded NC grade.
- Project grading: The student is expected to conduct presentation for the project. If absent for presentation, the student will be deemed as absent for the evaluative and will be awarded NC grade.

9. Make-up Policy

- Make up will be granted only on genuine reasons and will be considered on a case by case basis. However, prior permission is a must. It should follow the official request procedure. No other mode of request shall be entertained.
- For medical cases, a certificate from the concerned physician from the Medical Centre must be produced (Subjected to Academic regulations).
- Make up policy is only applicable to Mid – Semester test and Comprehensive Examination. It is not applicable to quiz and project presentation.

10. Note

Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

Instructor-in-charge
Course No. CE G567